

Divit Singhania

India | divitsinghania05@gmail.com | +91-9977819000 | Portfolio | linkedin | github

Experience

Research Intern, VIT Bhopal University – Bhopal, India

May 2024 – June 2024

- Developed MolSpectra, a molecular visualization tool from scratch using PyQt with 3D rendering via GLEW
- Integrated Open Babel, Gaussian, GAMESS, NWChem, and MOPAC for seamless quantum chemistry workflows
- Automated pipeline from molecular structure creation to quantum chemistry calculations
- Conducted literature review, data analysis, and contributed to technical report preparation for the SERB-sponsored CRG Research Project (CRG/20221002761)

Projects

NeuroSense

Jan 2026 – Current

- Designed an AI cognitive assessment platform detecting early Alzheimer's using multimodal inputs.
- Applied speech processing and NLP pipelines extracting linguistic and recall features.
- Created automated cognitive scoring using semantic similarity, recall accuracy, and coherence.
- Deployed backend modules for transcription analysis, feature extraction, and screening reports.

Minesweeper AI

Jul 2025 – Sep 2025

- Designed an interactive PyQt6 Minesweeper game with integrated AI, supporting two gameplay modes (manual + AI).
- Built a multi-head convolutional neural network trained on 50k+ board states for action, heatmap, and reasoning predictions.
- Generated structured training datasets (.npz, .csv, json) with 100k+ per-move snapshots for supervised learning.
- Developed a hybrid rule-based plus model-based decision system, improving AI gameplay accuracy by 35%.

FinRisk

Feb 2025 – Mar 2025

- Built an end-to-end machine learning model to predict loan default risk with 87% accuracy (AUC 0.91) using Python and scikit-learn.
- Deployed the solution as a Flask API on Hugging Face Spaces with Docker, serving 100+ test cases efficiently.
- Constructed robust preprocessing pipelines with 15+ financial features, including scaling and encoding.
- Delivered detailed probability scores, adaptive thresholds, and risk categorization, reducing manual analysis time by 50%.

Skills

Programming: Python, C++, Java

Machine Learning & Deep Learning: Supervised Learning, Feature Engineering, Model Evaluation, Cross Validation, CNNs, Transformers, TensorFlow, PyTorch, Keras

Data & NLP: NumPy, Pandas, Scikit-learn, Natural Language Processing, Speech Processing, Semantic Similarity

Tools & Deployment: Flask, REST APIs, Docker, Git, Hugging Face Spaces, Model Deployment

Extra-Curricular

Team Lead – Developer Team, iOS Club, VIT Bhopal: Led mobile and web development initiatives, coordinated team tasks, organized technical events, and mentored members in collaborative development.

Education

VIT Bhopal University, B.Tech in Computer Science (AI & ML)

Sept 2023 – Current

- CGPA: 8.63/10.0